

TEST II (2026)

PART II (Requirements engineering)

Team number:6

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For this test you have to build teams. One answer per team is required.

Deadline: 13th April 2026

Give short answers

(for the multiple-choice questions, select YES or NO with a reference to a slide number with some very short explanation)

Topic 5

5.1 What are **goals of the analysis and definition** phase:

a) Description of the requirements from the user's point of view YES NO

Answer 5.1 a): YES **X** NO

Reason: Topic 5 Slides 2 & 12:

One of the main goals is the definition of requirements to the software product from the point of view of the user.

b) Analysis of the problem area YES NO

Answer 5.1 b): YES **X** NO

Reason: Topic 5 Slides 12:

Analysis of the problem area is also listed as one of the goals of the analysis and definition phase

c) To derive requirements without any interaction with the customer of the software project
YES NO

Answer 5.1 c): YES NO **X**

Reason: Topic 5 Slides 19 – 22:

The customer is involved in producing all planning phase documents and participates in all activities like preliminary requirements specification, glossary, cost estimation and project plan.

d) Provide a basis for deriving the software architecture in later phases YES NO

Answer 5.1 d): YES **X** NO

Reason: Topic 5 Slides 16:

Product definition serves as the basis for the design phase, where software architecture is derived.

e) Provide a basis for deriving test cases in later phases YES NO

Answer 5.1 e): YES **X** NO

Reason: Topic 5 Slides 37:

Validity requirements include preparation of acceptance test cases for later phases.

f) To use a cost estimation to decide if the project should be continued YES NO

Answer 5.1 f): YES **X** NO

Reason: Topic 5 Slides 15:

Cost estimation is used to make a decision on the continuation of the project.

g) To introduce main technical terms in a glossary for reuse in many other documents (e.g. in requirements, in product model, in test cases, in user manual, in user interface)

YES NO

Answer 5.1 g): YES **X** NO

Reason: Topic 5 Slides 25:

Glossary defines terms to assure a unified terminology in the application field that can be reused for the requirements specification, the user interface, the online help and the user manual.

h) To derive acceptance test cases YES NO

Answer 5.1 h): YES **X** NO

Reason: Topic 5 Slides 37:

The requirements specification includes validity requirements, which cover the preparation of acceptance test cases.

i) To derive unit test cases YES NO

Answer 5.1 i): YES NO **X**

Reason: Topic 5 Slides 37:

The analysis and definition phase only covers the preparation of acceptance test cases as part of the validity requirements. Unit test cases are derived during the implementation and testing phase.

5.2 Review

a. A review is a process in which a team of evaluators systematically analyses a software document.

YES NO

Answer 5.2 a): YES **X** NO

Reason: Topic 5 Slides 51:

A review is the inspection of any software document by a group of evaluators through static reading.

b. A review refers only to a requirements specification (not to other documents like software architectures).

YES NO

Answer 5.2 b): YES NO **X**

Reason: Topic 5 Slides 51:

A review applies to arbitrary software documents, not only to requirements specifications.

c. A review in software development is impossible without running software. YES NO

Answer 5.2 c): YES NO **X**

Reason: Topic 5 Slides 51 & 54:

A review is performed by static reading of a document only. No running software is needed.

5.3 Which statements about **use cases** are true:

a. A use case represents a fundamental subfunction of the system YES NO

Answer 5.3 a): YES **X** NO

Reason: Topic 5 Slide 63:

A use case represents a fundamental system subfunction/service that delivers value to the user and is realized by a sequence of interactions between the user and the system.

b. Each elementary function of the system is also a use case YES NO

Answer 5.3 b): YES NO **X**

Reason: Topic 5 Slides 47 / 63 / 75:

Not every small or technical function is a use case. A use case is a high-level, meaningful function that delivers value to the user and consists of a sequence of interactions. An elementary function is often just a step inside a bigger process.

c. Use cases correspond to objects of the problem area YES NO

Answer 5.3 c): YES NO **X**

Reason: Topic 5 Slide 63:

Use cases and objects are different concepts because use cases describe behavior (functions) and objects define structure (entities/classes).

d. Use cases and test cases are nearly the same YES NO

Answer 5.3 d): YES NO **X**

Reason: Topic 5 Slide 63:

Use cases define system functionality, while test cases evaluate correctness, hence they are not equivalent (different purpose).

e. A use case must have an observable value for the user YES NO

Answer 5.3 e): YES **X** NO

Reason: Topic 5 Slide 63:

A use case must provide an observable result/value to the user, otherwise it is meaningless.

f. A user is called an actor and each actor has access rights to all of the use cases YES NO

Answer 5.3 f): YES NO **X**

Reason: Topic 5 Slides 47 / 60:

Actors represent human users or external systems, but they have specific access rights, not access to all use cases.

Topic 6

Which of the following statements are true/false for product models?

a) Product models make textual requirements specification more precise

YES NO

Answer 6.a): YES **X** NO

Reason: Topic 6 Slides 11-12:

The goal of a product model is to make textual requirements specification more precise by using basic concepts.

b) Product models are developed in the phase “Analysis and Definition”

YES NO

Answer 6.b): YES **X** NO

Reason: Topic 6 Slides 9-1:

The Definition phase includes the product model.

c) Product models are abstractions of systems to be developed

YES NO

Answer 6.c): YES **X** NO

Reason: Topic 6 Slide 19:

Model: Idealised, abstract representation of (a part) of reality (original) — contains significant aspects of original; omits insignificant aspects.

d) Product models cannot be constructed after the implementation of the system.

YES NO

Answer 6.d): YES NO **X**

Reason: Topic 6 Slide 36:

There is also Reverse Engineering- the model can be built from the existing system, even after implementation.

e) There is not only one model of a software system - it depends on the view, i.e. there might be quite different models of the same system.

YES NO

Answer 6.e): YES **X** NO

Reason: Topic 6 Slides 28-32:

There is not “the” only one model- it depends on the view.

f) There should be only one model of a software system which fits in the best way to the system

YES NO

Answer 6.f): YES NO **X**

Reason: Topic 6 Slides 28-32:

There may be several possible models that could fit in the best way to the system.

g) Software developers working in coding (i.e. programming) do not need a model since they have in mind the whole program code

YES NO

Answer 6.g): YES NO **X**

Reason: Topic 6 Slides 22-27:

Because of the complexity of the system, models cannot be avoided.

Topic 8

8.1 How can an OOA model be developed?

a) To read the textual requirements specification carefully YES NO

Answer 8.1 a): YES **X** NO

Reason: Topic 8 Slides 17 –19 and 42:

The OOA model is derived from the requirements specification.

b) To look for terms of the problem area in the requirement specification which might be classes in a class diagram YES
NO

Answer 8.1 b): YES **X** NO

Reason: Topic 8 Slides 42:

Classes are derived by finding classes in the problem domain from the requirements specification.

c) To read the use cases carefully to look for terms of the problem area YES NO

Answer 8.1 c): YES **X** NO

Reason: Topic 8 Slides 117:

Use case descriptions indicate candidates of possible objects of the problem area.

d) To study the design / software architecture of the system YES NO

Answer 8.1 d): YES NO **X**

Reason: Topic 8 Slides 9,14-16:

OOA is part of the analysis phase(early), while design and software architecture is part of a later phase.

e) To read the program code

Answer 8.1 e): YES NO **X**

YES NO

Reason: Topic 8 Slides 14-16:

Program code belongs to Implementation phase which happens after Design phase, so is much later than OOA.

f) To look at the test cases

YES NO

Answer 8.1 f): YES NO **X**

YES NO

Reason: Topic 8 Slides 14-16:

Test cases are part of Testing phase which is after Implementation phase, so it does not belong to OOA.

8.2 Which statements are true for an object life cycle?

a. It is part of the dynamic model of an OOA model

YES NO

Answer 8.2 a): YES **X** NO

Reason: Topic 8 Slides 36:

The object life cycle is part of the dynamic model, which describes the behavior of the system.

b. The object life cycle should specify the behavior of an object during run time of the system by describing its states

YES NO

Answer 8.2 b): YES **X** NO

Reason: Topic 8 Slides 11 and 62:

The object life cycle specifies an object's runtime behavior by defining its states.

c. The object life cycle is a model that is based on the well-known state automat (state machine)

YES NO

Answer 8.2 c): YES **X** NO

Reason: Topic 8 Slides 27 and 63:

The object life cycle is based on the concept of a state machine (state automaton).

10. What is measured by function points in the function point method?

a. Run time of the system

YES NO

Answer 10 a): YES NO **X**

b. Size of external memory

YES NO

Answer 10 b): YES NO **X**

c. Size of object code YES NO

Answer 10 c): YES NO **X**

d. Size of the program code YES NO

Answer 10 d): YES NO **X**

e. Effort to develop a system YES NO

Answer 10 e): YES **X** NO

f. Size of requirements YES NO

Answer 10 f): YES **X** NO

Reason: Topic 10 Slides 31, 33, 37: The FP method measures the “size of requirements” from the requirements specification. This size is then transformed to MM (man months) via company tables, representing the effort to develop a system. FP does not measure runtime, memory, object code, or program code size directly.

11. What are main approaches to cost estimation:

a. Analogy method YES NO

Answer 11 a): YES **X** NO

b Functional testing YES NO

Answer 11 b): YES NO **X**

c. Structured analysis YES NO

Answer 11 c): YES NO **X**

d Multiplier method YES NO

Answer 11 d): YES **X** NO

Reason: Topic 10 Slide 18: The four basic approaches to cost estimation are: analogy method, multiplier method, weighting method, and percentage method. Functional testing and structured analysis are not cost estimation methods.

12. Which of the following statements are true/false concerning the function point method:

a) Influencing factors (like use of decentralized data) will change un-weighted function points to (final) function points.

YES

NO

Answer 12 a): YES **X** NO

Reason: Topic 10 Slides 36–38: Influencing factors adjust un-weighted FPs by $\pm 30\%$ to produce weighted (final) FPs via the formula $E3 = E2/100 + 0.7$.

b) For the application of the function point method we need a transformation table of function points to MM.

YES

NO

Answer 12 b): YES **X** NO

Reason: Topic 10 Slides 58–59: A transformation table from FP to MM is needed as step 6 of the method.

c) This transformation table is independent of the company that is using the method.

YES

NO

Answer 12 c): YES NO **X**

Reason: Topic 10 Slides 57, 60: This table is company-dependent, IBM and VW have different values because of differences in project nature and experience.

e) The transformation of function points to MM is a linear function.

YES

NO

Answer 12 e): YES NO **X**

Reason: Topic 10 Slides 59–60: The transformation is non-linear, costs grow non-linearly as FP increase, due to increasing communication overhead in complex systems.

Send your answer to:

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with email subject "TEST II, Team 1, 2 ..." (respective team number)